ARE 106

1. NO DISCUSSION SECTION
This week

- we start next week

② MT1 = Friday April 24

MT2 = Friday May 15

Final = MONDAY June 8

6:00 - 8:00

* SYLL is POSTED to CANVAS

→ May be an error on SYLL.

RANDOM VARIABLE

- a variable whose value is unknown until it is observed.
- a RV is NOT perfectly predictable beforehand

$\boxed{\text{Ex}}$ X = outcome of a ROLL of a single die

<u>Before Roll</u> → X = random var
= unknown

<u>After Roll</u> → X = NO LONGER RV
= KNOWN

<u>There are 2 TYPES of RVs</u>

① Discrete RVs

② <u>CONTINUOUS</u> RVs

① <u>DISCRETE</u> RV →

- RV can take on a FINITE NUMBER of outcomes
- OUTCOMES are COUNTABLE
- Finite outcomes

② CONTINUOUS RVS

- Takes on an INFINITE number of values
- OUTCOMES are NOT COUNTABLE
- INFINITE OUTCOMES

Formulas for EXPECTED VALUE (X)

① $E(X)$

② $Var(X)$

Ⓐ DISCRETE

Ⓑ CONTINUOUS

Ⓐ DISCRETE RVS

$f(x_i) = Prob(X = x_i)$

$\mu_x \longrightarrow E(X) = \sum_{i=1}^{N} x_i f(x_i)$

$$Var(X) = \sum_{i=1}^{N} (x_i - \bar{x})^2 f(x_i)$$

(B) CONTINUOUS RVS

$$\mu \rightarrow E(X) = \int_a^b x_i \cdot f(x_i)$$

σ = standard deviation

$\rightarrow \sigma = \sqrt{\sigma^2}$

$SD(X) = \sqrt{Var(X)}$

$$Var(X) = \int_a^b (x_i - \mu)^2 f(x_i)$$

ROLL a SINGLE DIE

$$Prob(X = x_i) = f(x_i)$$

CONSIDER OUTCOMES AND THEIR ASSOCIATED PROBABILITIES

$\rightarrow P(X=1) = f(1) = \frac{1}{6}$

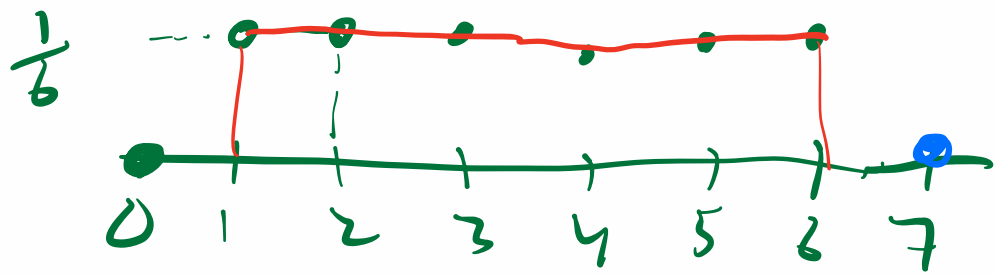
$P(X=2) = f(2) = \frac{1}{6}$

$\vdots$

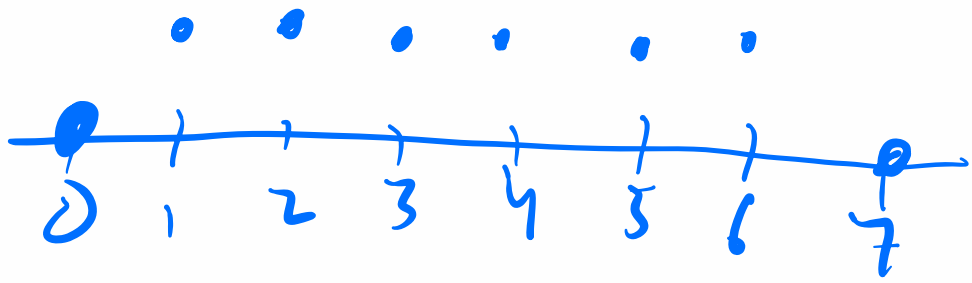
$$P(X=6) = f(6) = \frac{1}{6}$$

SUM OF ALL PROBABILITIES EQUALS $1 = 100\%$

Case 1: ROLL a SINGLE DIE



Case 2



Example - Calculate the mean & variance of outcomes from a single die,

AND, variance — $E(X)$, $Var(X)$

$$E(X) = \sum_{i=1}^{6} x_i F(x_i)$$

BILLIONS

292,000,000

$$= 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + \dots + 6\left(\frac{1}{6}\right)$$

$$E(X) = 3.5$$

100 rolls

$$E(X) = \frac{3 + 1 + 1 + 6 + 5 + 5 + 4}{100} \longrightarrow 3.5$$

Large number of Rolls